

Terraform

Monday, September 14, 2026 10:01 AM

30% theory and 70% hand on

Wrapper

You need to know the flow of how cloud works

You need to get the code as it is from terraform registry.

HIPAA (Medical Record) and PCI DSS Compliance

Data leak.

When you are creating infra using a code. I can scan the code for vulnerability also. Tflint checkov snyk.

Well secure architecture.

Developer once they build the code before it goes to deployment. They static code analysis for vulnerability testing.

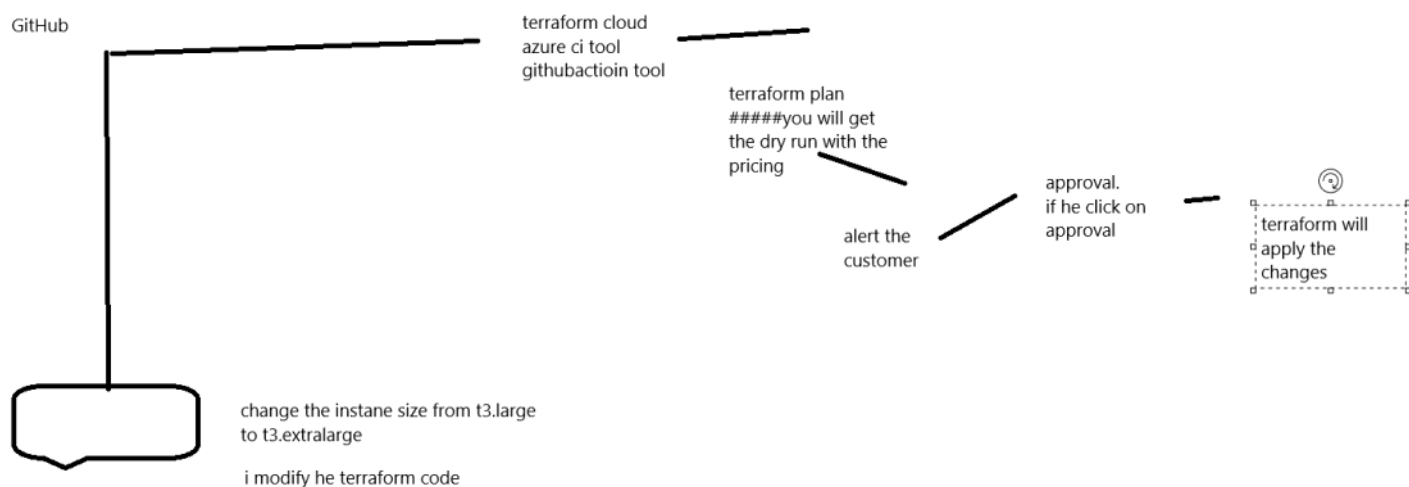
Who should create the document. The tool should create the documentation.

Terraform will automatically create the documentation keep the logs. It is perfect.

We are creating similar kind of infra. I will create a terraform template using the template I can create multiple infra in different region also.

I am running an ecommerce application. With 2 server with 4 core processor and 8GB of RAM for the past three months. Where we are selling cloths. I work up in the morning. Check application found that there are request which is failing in my site.

Scale up the application. I pay more on the cloud. I need to scale down. This also required automation.



Screen clipping taken: 9/14/2026 10:42 AM

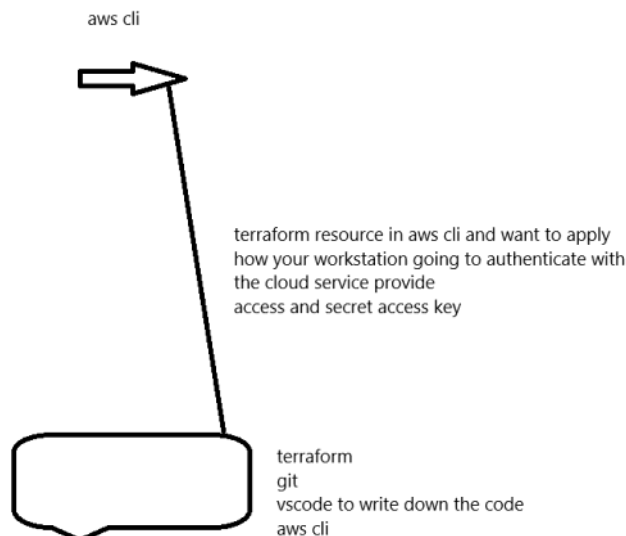
Stability: I created resources using terraform. Someone make changes in the resource from the cloud console. If you are managing your resource using terraform. Outside terraform this is called as drift.

Open port 8080.

I modify the sg make the change and apply it. This will be recorded in tfstate file.

Port 8080-8081

Terraform plan: this will give me the plan. What is there earlier 8080 and what is new 8081



Package installation

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Terraform

#different from different os

But working with terraform will be same .

Mac windows linux desktop mint ubuntu.

We will create everything in linux machine. Windows using terraform.

Choco package manager

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Choco package manager in windows

Yum apt

Terraform command

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I have created code to create vpc sg subnets ig custom rtb. Once I created code. All this code will be created inside a folder.

Terraform init

Terraform validate

Terraform plan

Terraform apply

Terraform destroy

Terraform fmt

Terraform graph

Configuration details to create resource in the cloud. Which cloud. Aws. It will create a .terraform folder inside it will download the plugin. To communicate with your cloud service provider.

Terraform plan: when I run this command on the current directory it will read all your manifest file or configuration file. Vpc subnet sg vm. Current state. Compare the manifest file with the cloud service provider and provide you a desired state.

Terraform apply. It will run on the current directory read all your manifest file compare it with cloud service provider specification. According to that if we say yes it will create the resource in the cloud--->it will store the information what has created inside a file called as terraform.tfstate

Instance size from t3 micro to t3 medium. Again it read all your manifest. Only the changes will be applied

If you lose your state file. S3 storage. Automatic backup. Remote state.

Terraform destroy: whatever resources is created it should terminate the same. Will read which file. Tfstate file.

Blocks

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Using terraform what we are going to do.
Create resources.

We want to create a vpc

1. When I try to create a vpc.

Output block

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We have created a template of terraform using the same developer can deploy there resources in the cloud.

1. Vpc.subnet,sg,custom rtb,internetgw, instances,loadbalancer.
2. Once the application is deployed in the cloud. They need to get the lb dns name using the same they can access the application.
3. We have not given access to aws console to the developers. Output block is the once your terraform script has completed you can display the output in your terminal

Calling block

Data and module block

Data block:

Anything which is being created outside terraform we will call the same using data block.

Like creating instance. We use an ami. I want to call that os image.

Data block.

Rds : using terraform we created the rds db. To login inside rds instance you required password.

Terraform resource creation

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1. We will create our vpc
2. Subnet
3. Custom rtb
4. Internetgw
5. Launch our instance in our custom vpc

Using aws console we can do

Directly start with terraform coding.

Aws console-then we start coding

1. Vpc
2. Subnet
3. Customrtb
4. Internet gw attach internet gw withy our vpc and custom rtb
5. Then we will launch an instance

Platform ready

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1. We know using console how to create the resource in aws cloud
2. We have terraform installed with plugin
3. Our workstation is being authentication to the aws cloud using access key and secret access key

We need to start coding.

1. Vpc
2. Subnet app and db subnet
3. Custom rtb
4. Ig
5. Custom rtb ---attach ig --app subnet
6. Then only we can launch our instance.

Rds

Vpc

Minimum two subnet they are also in two different data center az.

recap

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We learn about IAC and how it can help you to manage your resources in the cloud.

We learn about terraform workflow.

Terraform init: this is used to initialize the provider plugin

Terraform fmt: will do syntax validation

Terraform plan: this will compare all your manifest file with cloud service provider and give you an dry run desired state

Terraform apply: again will read all your manifest file give you desired state. Once you say yes it will create the state and store those information inside a file called as terraform.tfstate

Terraform destroy: will read your state file and destroy all the resources.

Then we create a vpc-subnet-ig-association custom route table and launch the vm in the app subnet and access the virtual machine.

Web_vpc: this is the reference block given to create the vpc. Using this reference block terraform store all the information about your resources inside state file.

Refrence block

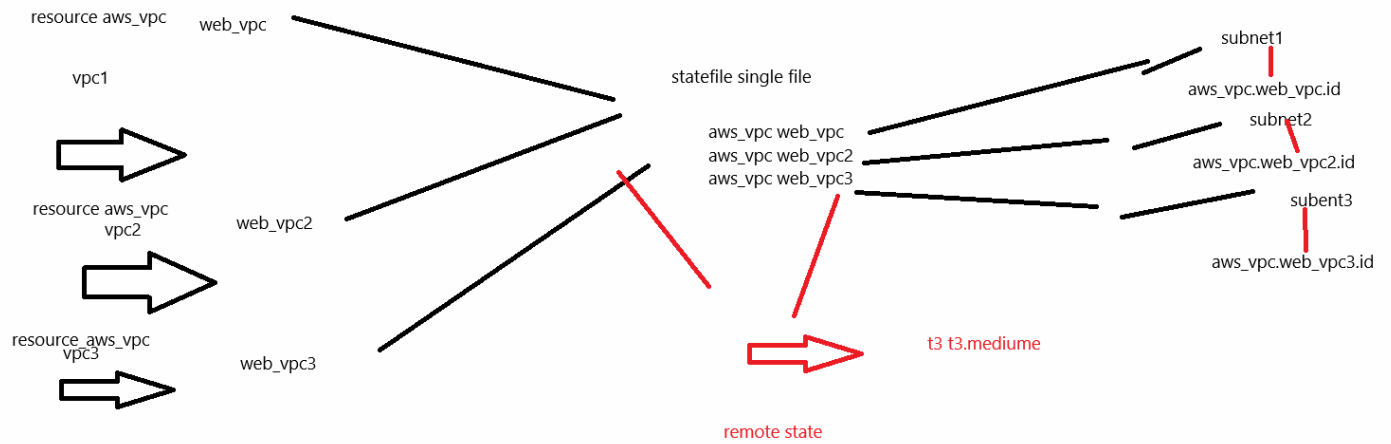
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The first purpose of the refrence block is to store the information about the resource inside a file tfstate file

2. Refrence block is also used to map another resources.

What is the two different purpose of this refrence block.

1. Store all the resource information inside your state file
2. When a mapping resource like subnet I am creating against my vpc I will get the mapping value from the state file.



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Variables

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They are called as input variables.

When you declare a value and you want to reuse it multiple times.

Walmart they have more than 10 thousand application.

It happen using terraform.

Every time I cannot create the terraform script with the same purpose.

We try to find out what are the similar kind of infra we need for common application according to that we created terraform script.

Instance size

Region

Vpc cidr block

We created an variables file using this files. Modify the variables and do terraform apply.

You declare a variables to call the variables we will use var.nameofthevariables

Local

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With variables we can pass a single value.

Suppose I want a meaning name to my project. Using two variables.

I want to concatenate multiple variables.

Named expression should be always in key value format

Gopal

Anger = staring

Fear = stretched

Happiness= smile

Sadness=dropped lips

Purpose of local

1. Concatination
2. Named expression

Object variables

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When you want to loop around with multiple values.

Duplicacy of the block

One single resource block can create similar kind of resources with different values.

Loops

For loops

We will use something called as object variables in terraform. It is kind of dictionary. Where we define nested values

Student database

Student1

Age

Subject

Address

Student2

Age

Subject

Address

Student3

Age

Subject

Address

We will create an internet gw attach it with my vpc

Create an custom rtb with this custom rtb we are going to attach our ig and assoicate our subnet public subnet assoicate

Output block

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Once we create the resource I every time go to aws console and check the resources it is create or not
Once the resource get created we want to display the resource information in the terminal only
There are developer who use our terraform script to deploy the application once deployed
To access the application they can use the public ip or lb dns name
Using the output block we can display only the required information
We have created vpc public subnet private subnet internet gw and route table.

```
"aws_vpc" "web_vpc"
```

```
"aws_subnet" "web_subnet"
```

```
output "public_subnet" {  
  description = "ids of all public subnet"  
  value = {  
    for name, subnet in aws_subnet.web_subnet :  
      name => subnet.id  
  }  
}
```

Value = aws_subnet.web_subnet.id : this will get only one single subnet where as we have multiple subnet

If you want to get one single subnet

```
for name, subnet in aws_subnet.web_subnet
```

What does this for expression do

#it will loop through every subnet inside aws_subnet.websubnet

#there are two variables

Name

Subnet

Name if the key

Subnet1

Subnet2

Subnet3

#for name

#when it do the first iteration

Name = "subnet2"

Subnet = aws.web_subnet["subnet1"]

Name = "subnet2"

Subnet = aws.web_subnet["subnet2"]

Name => subnet.id

This will create a map

Key => map

Name => subnet.id

It will create a map by looping through every resource in aws_subnet.web_subnet using the subnet

for_each

```
Subnets={  
Subnet1=subnet1id  
Subnet1=subnet1id  
Subnet1=subnet1id  
}
```

#in conceptual way
Get the name
Get the subnet object
Get the subnet id
Add
Name -> subnet id

For : start the loop
Name: key of each item
Subnet value resource object
: start the result expression
=> maps key to value

For every subnet
Take its name
Take its id
Create
Name+id

recap

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We learn about

Resource block.

We created resources using terraform.

Resource name of the resource reference block.

1. To refer the resource
2. Using the reference it store the information inside your state file.

Vpc-subnet we use the reference block

Variables: using variables we can do reusability of the code.

Input variables

String: will accept only single value

Object variables is an data structure.

How to call this object inside resource block.

Using for_each

Each.key.name of the variables

Each.value name of the variables.

We are using for each loop so that we can make our code clean also we can easily do the repetitive task.

With two values

We want to create multiple subnet but this subnet need to created in multiple az.

In our subnet we want launch instance of different size

Local block

When you want to concatenate multiple variables

Named expression

Output: display the value in the terraform console

Graph

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It will show the mapping of all the resources in a visualization platform.

Vm

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#we can create an virtual machine

1. And startup script boot strap script
2. You are creating an linux vm we can provide a shell script wile launching the instance it should run those script
3. Windows machine ps1. powershell script
2. Before we launch our instance.
3. We call ami which is being created outside terraform

Data block

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In terraform a data block or data source is used to fetch information from resource that already exist in your infra or are managed outside of your current configuration.

In resource block we tell to create create update or destroy infrastructure. A data block will do read only operation.

It will query the cloud like aws retrieve the details in real time and make those attribute available to other parts of your code

Dynamic frequently changing data also

Here we are using data block to call our ubuntu image as the script is apt so we will use ubuntu image

Security group

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This need to create without that our infra will not be secured at all
This security need to be attached where. Vpc or you can attach with your with subnet also.

Data center

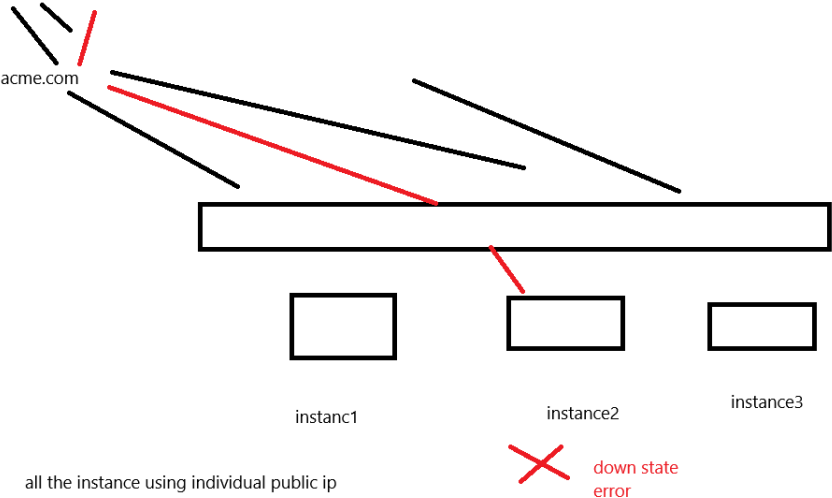
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We have three different data center.
Currently we deployment a single instance in the data center. Ha or resilient architecture.
How to create ha

When I want to create one more instance in another data center.
I also need to do input of subnet. According to the subnet it should create those instance.
Subnet is attached to your data center

Subnet1 t3.micro instance
Subnet2 t3.medium
Subnet3: t3.small

We need to loop around multiple values
1. Instance size
2. subnet



- 1. distribute the traffic to multiple instance. lb
- 2. lb will send the traffic to multiple instance using round robin fashion
- 3. what step we can do so that this error will not come.
 - it do the health checkup
 - keep on pining my server
 - if it do not recive response it should stop sending the traffic.

Lb

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1. When I want to create lb
 - a. Name of the load balancer
 - b. Internet facing
 - c. Add the vpc
 - d. Select two az
 - e. Separate sg for lb: 80 443
 - f. Instance sg: 22 80 443
 - g. Fwd to target group - group of instances

#we need to create lb first or the target group.

Target-group:

Health threshold: 2

Interval 30 second

Once of the machine goes down. Lb will ping your server multiple times 2 times every 30 second it will ping your server 1 minues it will start sending the traffice to that instance.

Unhealth 4

Interval is 30 second

Ping your server 4 times 2 min. if your server do not respond it should stop sending the traffic.

We will first create a security group

Target group

Lb

Acme.com - A-Record- gopal-project-dev-alb-1181065731.us-east-1.elb.amazonaws.com

Recap

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We create a vpc

Inside the vpc we created multiple subnet using object variables create the subnet in different az

We create ig and custom rtb attach the custom rtb with public subnet and internet gw routing

We create the lb

Target group

Listener in the tg with health check up

Remote state

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What is the purpose of state file?

1. Keep the logs
2. Current state of infrastructure
3. Avoid delicacy

No

Code will be different state file is like an logs

When we do terraform apply the state file goes inside tf state

When we do terraform destroy state file goes inside terraform.tfstate.backup

What I do in Monday and what I do today there will be no logs available

Previous state file should be available a month also I should have the state file.

As it is your local machine it is getting overwritten

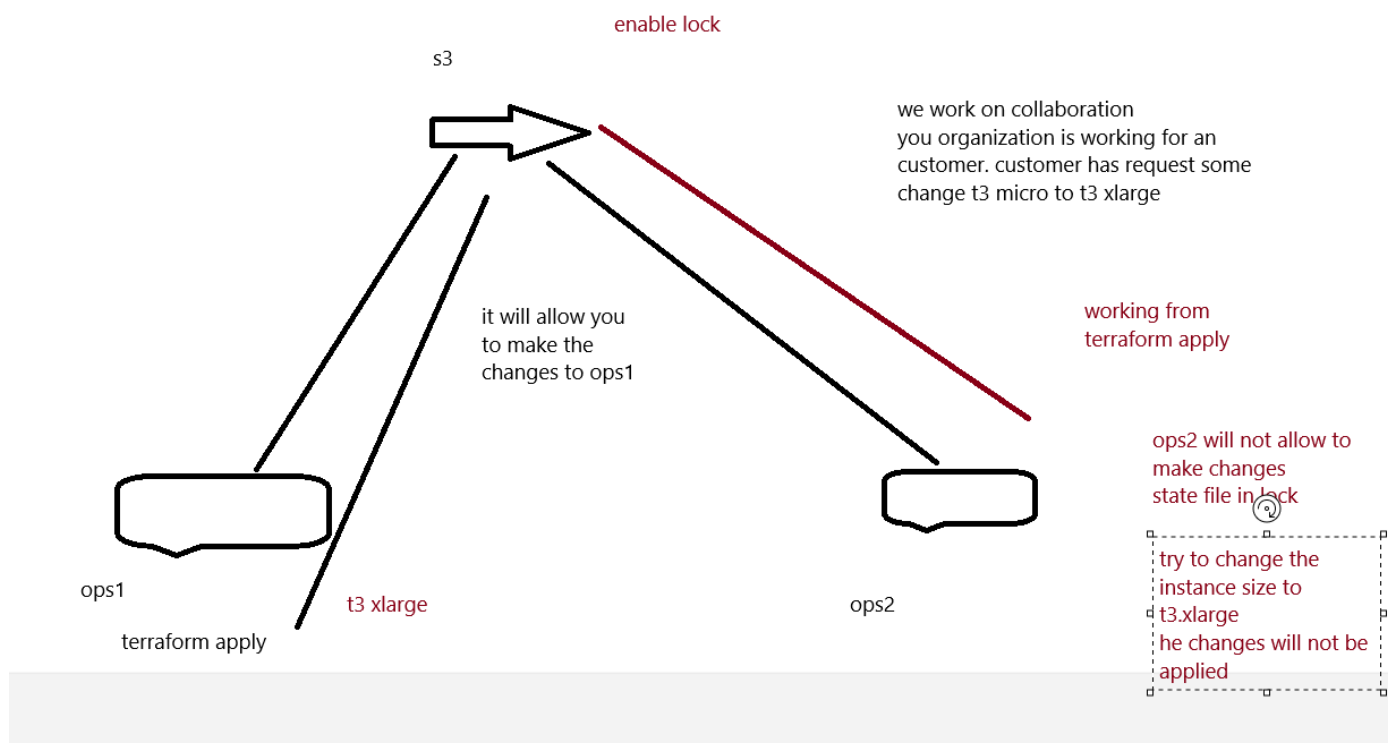
You need an storage if I upload the same file multiple times then also the file will not be overwritten

enable versioning in s3 bucket.

Filename: timestamp

Blob storage

Storage account



Screen clipping taken: 9/17/2026 10:03 AM

With the s3 bucket you cannot see it .

Use hashicorp terraform cloud.

You will give rbac to the project of yours

Version control system

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Terraform remote state we manage your state and do the versioning

Similar to that we are writing code. This code need version control system to manage.

We created a well defined template. Using the template we can create resources.

Dev test or prod environment.

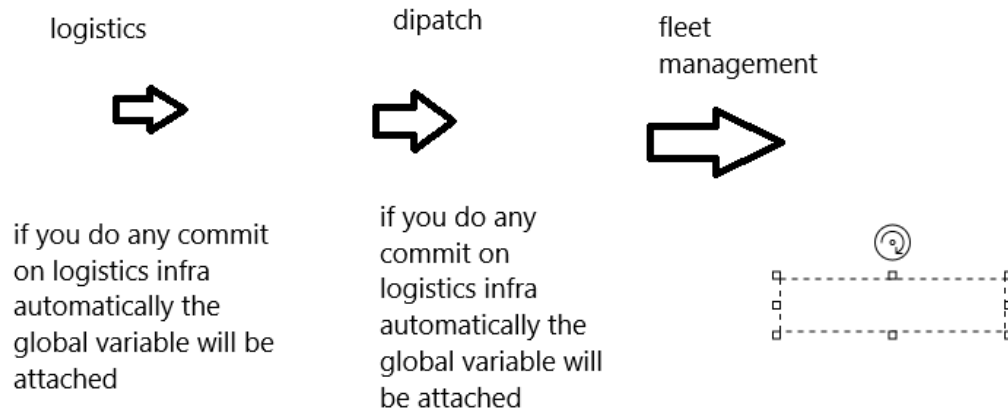
Same code we do the versioning and branching and create all the environment

Terraform cloud it don't have the versioning system

You need to pass the code to the terraform cloud using vcs

Git is vcs

user.name
user.emailid



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Git workflow

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#i have code in one of the folder IAC

Journey start with

##how git is going to manage the code

Git init: which track the code git add git commit your first version got created you upload the code in a remote repo github azure repo

Git is your local client inside your laptop

Github remote repo



Screen clipping taken: 9/17/2026 12:03 PM

Terraform cloud

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We will see how to upload our terraform code we have build which create resource in the cloud
How to upload the same in git hub and integrate with terraform cloud.

recap

Friday, September 18, 2026 9:43 AM

Terraform is provisioning tool

Configuration management: ansible chef puppet ansible to do the patching.

Terraform cannot keep the logs. machine

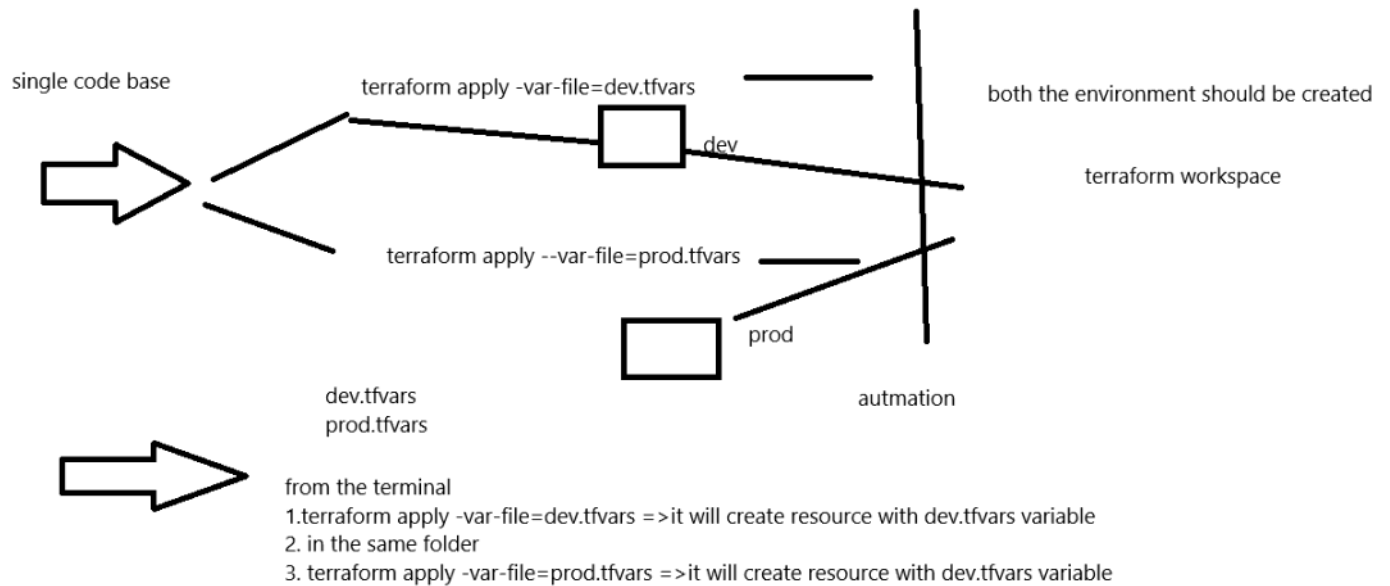
Terraform workspace

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Code base the folder where we have stored all the terraform manifest file.
What are the reusable component inside the code base.

To do the same we need to first declare our variables inside a file the extension of the file is .tfvars

When I share my template which create some infra with another team member.
Another team member work on with your template. 172.31 project different



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Variables.

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Terraform apply

If you do not pass on the variables file first precedence it will take the default value from the variables declaration file

Terraform apply --var-file=dev.tfvars

It will over write all the declared variables block default value with dev.tfvars

Terraform workspace new dev

Creating a dev folder and copying all your code from the root folder to the dev folder terraform apply --var-file=dev.tfvars

Creating a prod folder and copying all your code from the root folder to the prod folder terraform apply --var-file=prod.tfvars

Modules

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Terraform workspace.

This environment can be created using passing the variables files.

We increase the reusability of the code. By deriving the code from a single base.

Again this help you to do modularization of the code.

You are working for a client the client is in overseas location. Europe.

He has build terraform code as per the organization standard. Banking sector. Stringent in the law.

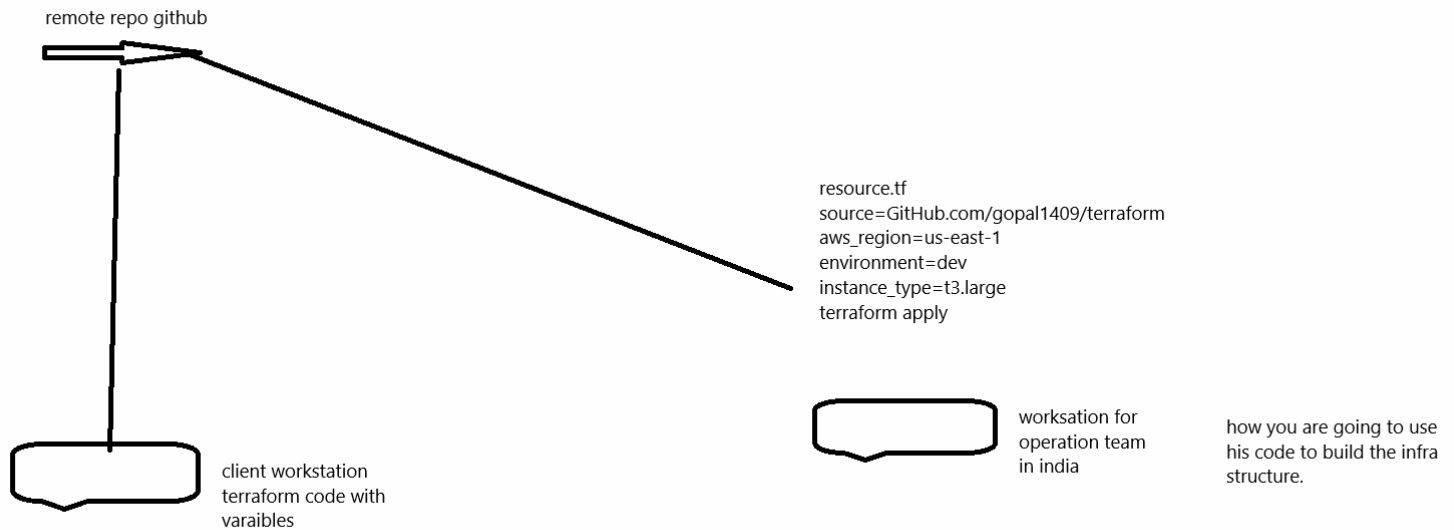
Everything should be compliance as per government norms.

Your job is to build infra for them using terraform.

The skeleton has be created by the team in europe. By changing some variables.

How you will get the main code.

They will upload the code in github.



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Module reusability

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When we are creating terraform code. And want to convert the same into modules.
I need to ensure that the code should be easily understandable.

As per our terraform configuration we can refactor the project into four reusable modules

1. Networking: vpc public subnet private subnet internet gw route table
2. Security group - web sg alb sg
3. Compute: ubuntu ami data block ec2 instance app.sh
4. Load balancer : alb tg listener target

First thing we will create directory structure and upload the same in github